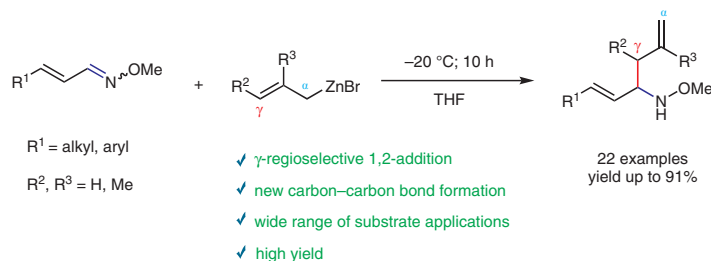


Highly γ -Regioselective 1,2-Addition of α,β -Unsaturated Oxime Ethers with Allylzinc Bromides: A Straightforward Approach for the Synthesis of Homoallylic Amines

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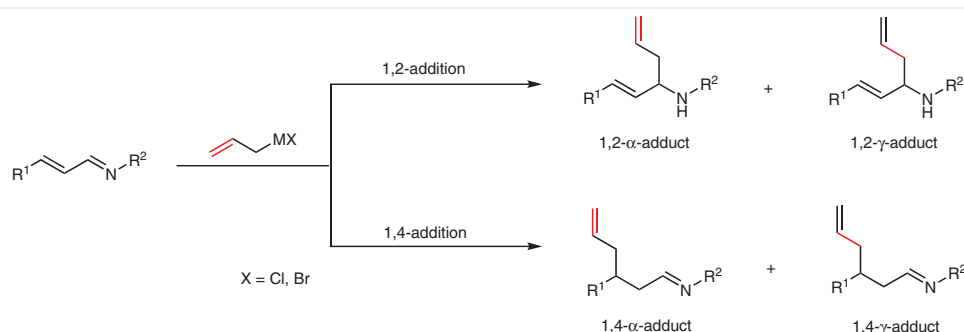
Abstract A highly regioselective reaction between allylzinc bromide reagents and α,β -unsaturated oxime ethers for the one-step synthesis of the homoallylic amines is reported. This process is a regioselective 1,2-addition reaction providing a new γ -position with carbon–carbon bond formation. Furthermore, the reaction substrates are widely applicable and can be produced in a high yield.

Key words regioselectivity, homoallylic amine, α,β -unsaturated oxime ether, organozinc reagent, 1,2-addition

The conjugated addition reaction of α,β -unsaturated carbonyl compounds and their equivalents with organometallic reagents is a basic method for preparing C–C bonds.¹ Since nitrogen atom play an important role in a large number of bioactive molecules,² the regioselective conjugate addition of α,β -unsaturated imines has been an important area of concern for organic chemists.³ Among them, because allyl groups are important universal handles for organic synthesis,⁴ the addition reaction of allylmethyl re-

agents with α,β -unsaturated imines has also become the focus of many researchers. However, due to the effects of electrons, steric hindrance,⁵ and other factors, as well as the specificity of allyl groups,⁶ there are many competing results for the conjugated allylation of α,β -unsaturated imines (Scheme 1). Therefore, establishing a highly regioselective addition of an allyl group to an α,β -unsaturated imine is a huge challenge.

Homoallylic amines are an important class of synthetic intermediates and substructures in medicinal chemistry⁷ and can be widely used to synthesize large quantities of nitrogen-containing natural products, heterocyclic compounds, and biologically active molecules.^{2b,8} In the past few decades, organic chemists have made extensive efforts to develop effective preparation of such compounds. Generally, homoallylic amines are prepared either by addition of organometallic reagents (such as indium,^{7h,9} magnesium,^{7k,10} samarium¹¹) to imines or by nucleophilic addition of allyl reagents (such as allylsilane,¹² allyltin,¹³ allylboron,¹⁴ allyltitanium, or allylgermane¹⁵) to imines in the presence of acid catalysts. However, the latter method generally involves a three-component reaction and cannot be carried

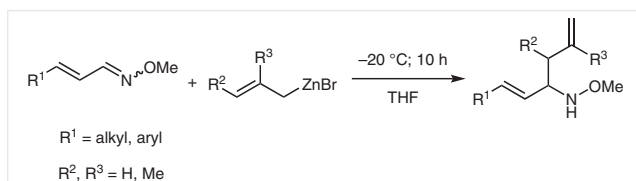


Scheme 1 Selective competitive addition of allyl metal reagents to α,β -unsaturated imines

out in one pot due to the water and amine present during the formation of the imide, which tends to decompose or deactivate the Lewis acids.¹⁶ Therefore, the preparation of homoallylic amine by the addition of an organometallic reagent to an imine is now the mainstream of research. In fact, due to the low electrophilicity of the imine, its allylation process requires the use of more reactive reagents. As a member of the imine derivatives, oxime and oxime ethers are more likely to produce aziridine by-products due to the increased acidity of the α -protons,¹⁷ so that it is more difficult for them to undergo a nucleophilic addition reaction. To date, only effective allylation processes of a small number of oxime ethers derived from glyoxylic acid and 2-pyridinecarboxaldehyde have been reported.^{9f,18} Therefore, the use of a simple strategy to achieve allylation of oxime ethers will be an interesting topic.

Zinc is a relatively non-toxic and inexpensive metal, and it is a trace element necessary for human and all animal and plant life activities. Since its discovery more than 150 years ago,¹⁹ organozinc reagents have been greatly limited in their application to organic synthesis due to their low reactivity. Compared to Grignard reagents and organolithium reagents, it has higher functional group compatibility in organic reactions.²⁰ Nowadays, organozinc reagents have become one of the most important organometallic reagents in organic synthesis.²¹ For a long time in the past, we have achieved some research results²² in the application of zinc to organic synthesis. Therefore, we have a good understanding of zinc reagents. Based on the above research, we try to further expand the application of zinc in the synthesis of homoallylic amines.^{9e,23}

Herein, we describe a basic process for the reaction of an α,β -unsaturated oxime ether with an allylzinc bromide reagent under mild conditions to the one-step synthesis of the homoallylic amine (Scheme 2). The reaction produces a unique γ -targeted 1,2-addition product with high regioselectivity and in high yield.



Scheme 2 Synthesis of homoallylic amines from α,β -unsaturated oxime ethers and allylzinc bromides

In the initial study, we used (2*E*)-cinnamaldehyde *O*-methyl oxime (**1a**) and allylzinc bromide (**1b**) as template substrates to explore the optimal conditions for this reaction (Table 1). By adjusting the material ratio between the two reactants at room temperature, it was found that when the allylzinc bromide (**1b**) was 2 equivalents, the yield of the obtained homoallylic amine **1c** was 51%. When the allylzinc bromide (**1b**) is 1 equivalent or 1.2 equivalents, the

reaction outcome is unsatisfactory (Table 1, entries 1–4). In the case where the optimum material ratio was determined, we optimized the temperature of the reaction (entries 5–9). We have found that the reaction yield gradually increases as the temperature gradually decreases from room temperature. The reaction yield was as high as 68% when the temperature was lowered to $-20\text{ }^\circ\text{C}$ (entry 8). It is worth noting that when the temperature was lowered to $-25\text{ }^\circ\text{C}$, **1a** in the system is not completely converted, and the yield dropped again to 61% (entry 9). From this we can confirm that the optimum reaction temperature for this reaction is $-20\text{ }^\circ\text{C}$. Excitingly, when we extended the reaction time to 10 hours, the reaction yield could reach 87% (entry 10), and the yield was slightly reduced by continuing to extend the time to 12 hours (entry 11). Taken together, the above experimental facts indicate that the optimum conditions for the reaction are: 1:2 molar ratio of **1a** to **1b** with a reaction time of 10 hours at $-20\text{ }^\circ\text{C}$.

Table 1 Optimization of Reaction Conditions^a

Entry	1a / 1b ^b	Temp ($^\circ\text{C}$)	Time (h)	Yield (%) ^c
1	1:1	25	7	24
2	1:1.2	25	7	27
3	1:1.5	25	7	49
4	1:2	25	7	51
5	1:2	0	7	52
6	1:2	-10	7	64
7	1:2	-15	7	66
8	1:2	-20	7	68
9 ^d	1:2	-25	7	61
10	1:2	-20	10	87
11	1:2	-20	12	83

^a Reaction conditions: **1a** (0.5 mmol) was added to a THF solution of **1b** and allowed to react under N_2 .

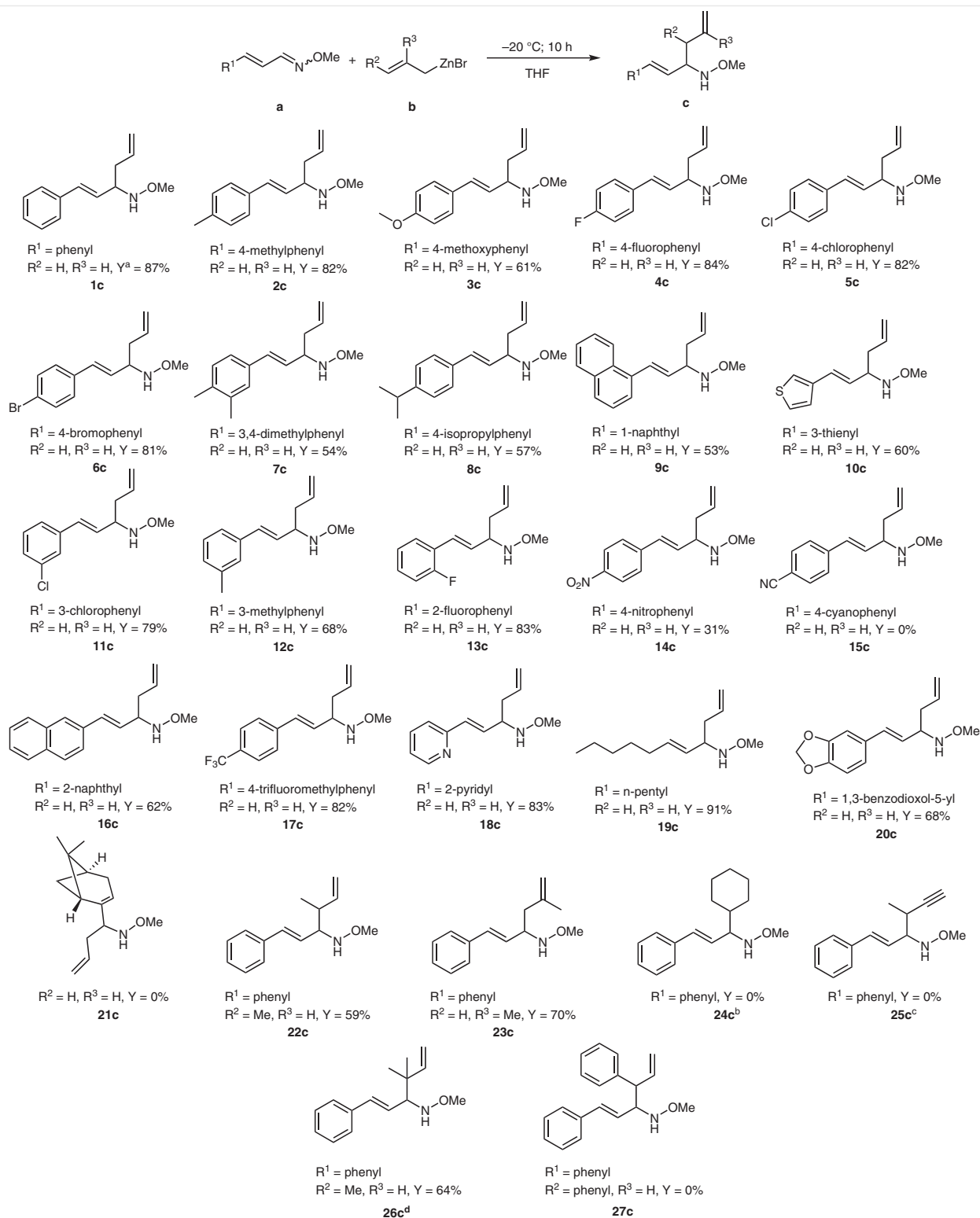
^b From a 0.2 mol/L stock solution of **1b** in THF, 2.5 mL, 3 mL, 3.75 mL, and 5 mL, respectively, was used to increase the material ratio of **1b**.

^c Isolated yield based on **1a** after basic alumina chromatography.

^d A small amount of **1a** remained as residue.

With optimized reaction conditions in hand, we next used a variety of different α,β -unsaturated oxime ethers and allylzinc bromide reagents to explore the substrate suitability of this reaction. The results of this study are shown in Scheme 3.

As shown in Scheme 3, the aryl-substituted α,β -unsaturated oxime ether reacts with allylzinc bromide providing very good reaction results (Scheme 3, $\rightarrow$ **1c–18c**, **20c**).



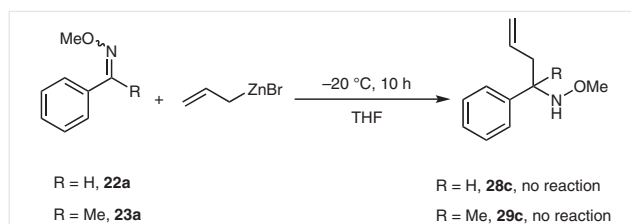
Scheme 3 Substrate range for the reaction of α,β -unsaturated oxime ether with zinc reagent. *Reagents and conditions:* **a** (0.5 mmol) was added to a THF solution of **b** (2 equiv) at $-20\text{ }^\circ\text{C}$ and allowed to react for 10 hours under N_2 atmosphere. ^a Isolated yield based on **a** after basic alumina chromatography. ^b Cyclohexyl iodide was used in place of allylzinc bromide. ^c Zinc reagent of 1-bromo-2-butyne was used in place of allylzinc bromide. ^d The reaction was carried out for 48 h at r.t.

Among them, whether it is an electron-rich phenyl-substituted (Scheme 3, $\rightarrow$ **2c–3c**, **7c–8c**, **12c**) or electron-deficient phenyl-substituted (Scheme 3, $\rightarrow$ **4c–6c**, **11c**, **13c**, **17c**) α,β -unsaturated oxime ether, a moderate to high yield was obtained. When the fused ring (Scheme 3, $\rightarrow$ **9c**, **16c**, **20c**) or heterocyclic (Scheme 3, $\rightarrow$ **10c**, **18c**) substituted α,β -unsaturated oxime ether is reacted with allylzinc bromide, the corresponding homoallylic amine can also be obtained in a good yield. Even some special functional groups such as nitro groups (Scheme 3, $\rightarrow$ **14c**) are allowed to participate in this reaction with a yield of 31%. Unfortunately, this reaction does not afford the product when a cyano-substituted (Scheme 3, $\rightarrow$ **15c**) α,β -unsaturated oxime ether was used as a substrate because too complex reaction products were formed. Later, we explored the effect of alkyl-substituted unsaturated oxime ethers on this reaction. We found that the effect is even more exciting when using long-chain alkyl groups in the reaction, with yields as high as 91% (Scheme 3, $\rightarrow$ **19c**). When a structure in which a cyclic alkyl group substituted with a steric hindrance is added to the reaction, the ideal reaction product cannot be obtained (Scheme 3, $\rightarrow$ **21c**), which is also expected. In order to determine whether the α - or γ -position attacked the unsaturated oxime ether, we used an allyl zinc bromide reagent with different substituents to react with **1a**; the products **22c** and **26c** were finally obtained in moderate yields. From this we can confirm that allylzinc bromide is a γ -site attacking α,β -unsaturated oxime ether to form a homoallylic amine. When cinnamylzinc bromide was used instead of allylzinc bromide to participate in the reaction, the reaction was very messy and no product was formed (Scheme 3, $\rightarrow$ **27c**). Finally, we used some non-allylic organic zinc reagents to expand this reaction, but the results were not very satisfactory (Scheme 3, $\rightarrow$ **24c–25c**). Among them, cyclohexylzinc iodide did not produce the required product **24c** because of the steric hindrance.

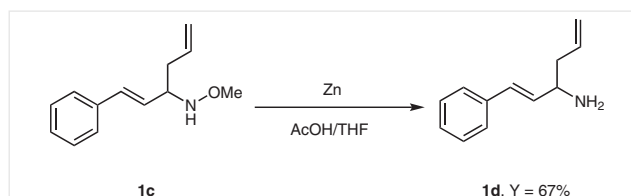
Consistent with the previous works reported by Wolan,²⁴ the reaction of simple oxime ether with allylzinc reagent does not take place (Scheme 4).

Finally, the elimination of the OMe through the reductive cleavage of N–O bond by metal zinc in product **1c** could afford the homoallylic amine **1d** (Scheme 5).

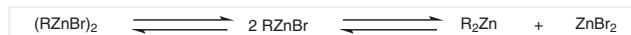
It is a complex mixture due to the presence of Schlenk equilibrium²⁵ in the allylzinc bromide solution (Scheme 6). Taking the allylzinc bromide reagent as an example, based



Scheme 4 Reaction of simple oxime ether without alkene group with allylzinc bromide



Scheme 5 Conversion of the product to afford a homoallylic primary amine

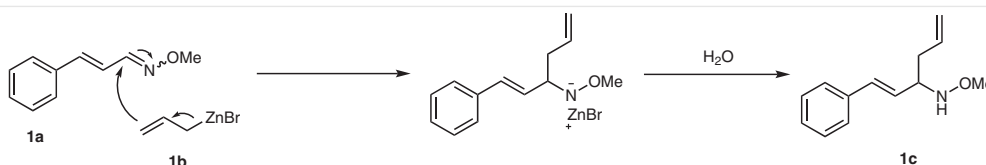


Scheme 6 The Schlenk equilibrium in the allylzinc bromide solution

on the previous experimental results, we speculated a possible reaction pathway as shown in Scheme 7.

In summary, we have demonstrated a highly γ -regioselective 1,2-addition reaction that occurs between an α,β -unsaturated oxime ether and an allylzinc bromide reagent. By this method, the homoallylic amines can be synthesized directly. The reaction has the advantages of simple conditions, single product structure, wide reaction substrate, and high yield. This work is another supplement to the selective addition of organozinc reagents to unsaturated imines.

THF was distilled from Na/benzophenone under N_2 . All reactions were conducted under a N_2 atmosphere. Metallic Zn and all solvents were purchased from commercial source, and used without further purification. Flash column chromatography was carried out on Macklin basic alumina (200–300 mesh) and Merck silica gel (300–400 mesh). ^1H and ^{13}C NMR spectra were recorded on a Varian-Inova 400 spectrometer. Solvent for NMR is CDCl_3 , unless the otherwise noted. Chemical shifts are reported in delta (δ) units in parts per million (ppm) relative to the singlet (0 ppm) for TMS. Data are reported as follows: chemical shift, multiplicity (standard abbreviations), coupling constants (Hz), and integration. ^{13}C NMR spectra were recorded on 100 MHz. Chemical shifts are reported in parts per million relative to



Scheme 7 Possible pathway of the reaction

the central line of the multiplet at 77.16 ppm for CDCl_3 . High-resolution mass spectra were obtained with a GCT-TOF instrument. All chemicals were purchased from Aldrich, Alfa, or Acros chemical company and used thus, without further purification. Petroleum ether (PE) used refers to the 60–90 °C boiling fraction.

Organozinc Bromides; General Procedure

Br_2 or I_2 (1 mmol) and Zn powder (72 mg, 1.1 mmol) were added to anhyd THF (5 mL) under N_2 at r.t. The mixture was stirred for about 30 min and during this time the Zn powder had disappeared. Stirring was continued for 1 h to complete the preparation of the organozinc reagent.

α,β -Unsaturated Oxime Ethers 1a–21a

For the preparation of α,β -unsaturated oxime ethers **1a–21a**, see the Supporting Information. The analytical and spectral data of **1a–21a** are listed below.

(2E)-Cinnamaldehyde O-Methyl Oxime (1a)²⁶

Colorless liquid; yield: 418 mg (52%).

^1H NMR (400 MHz, CDCl_3): δ = 7.86 (dd, J = 8.0, 1.1 Hz, 1 H), 7.43 (dt, J = 3.1, 1.9 Hz, 2 H), 7.37–7.26 (m, 3 H), 6.86–6.75 (m, 2 H), 3.92 (s, 3 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 150.58, 138.60, 136.10, 128.91, 126.99, 121.98, 62.03.

HRMS (EI+): m/z calcd for $\text{C}_{10}\text{H}_{11}\text{NO}$ ($M + \text{H}$)⁺: 162.0919; found: 162.0906.

(2E)-3-(p-Tolyl)acrylaldehyde O-Methyl Oxime (2a)²⁷

Colorless liquid; yield: 350 mg (40%).

^1H NMR (400 MHz, CDCl_3): δ = 7.85 (dd, J = 6.6, 2.5 Hz, 1 H), 7.40–7.32 (m, 2 H), 7.16 (dd, J = 8.9, 4.5 Hz, 2 H), 6.83–6.70 (m, 2 H), 3.93 (d, J = 18.0 Hz, 3 H), 2.34 (s, 3 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 150.80, 148.31, 139.07, 138.70, 133.38, 129.65, 127.58, 126.96, 120.97, 115.36, 62.00, 21.51, 21.46.

HRMS (EI+): m/z calcd for $\text{C}_{11}\text{H}_{13}\text{NO}$ ($M + \text{H}$)⁺: 176.1075; found: 176.1081.

(2E)-3-(4-Methoxyphenyl)acrylaldehyde O-Methyl Oxime (3a)²⁷

White solid; yield: 410 mg (43%); mp 86.2–87.1 °C.

^1H NMR (400 MHz, CDCl_3): δ = 7.88–7.80 (m, 1 H), 7.46–7.34 (m, 2 H), 6.92–6.84 (m, 2 H), 6.78–6.63 (m, 2 H), 3.93 (d, J = 18.4 Hz, 3 H), 3.81 (s, 3 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 160.31, 150.91, 138.35, 128.93, 128.41, 119.75, 114.37, 61.94, 55.45.

HRMS (EI+): m/z calcd for $\text{C}_{11}\text{H}_{13}\text{NO}_2$ ($M + \text{H}$)⁺: 192.1025; found: 192.1024.

(2E)-3-(4-Fluorophenyl)acrylaldehyde O-Methyl Oxime (4a)²⁷

White solid; yield: 304 mg (34%); mp 57.4–57.9 °C.

^1H NMR (400 MHz, CDCl_3): δ = 7.84 (dd, J = 6.0, 3.1 Hz, 1 H), 7.50–7.37 (m, 2 H), 7.08–7.00 (m, 2 H), 6.82–6.71 (m, 2 H), 3.94 (d, J = 16.5 Hz, 3 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 163.09 ($^1J_{\text{CF}}$ = 250.2 Hz), 150.41, 147.88, 138.64, 137.29, 132.39, 132.35, 129.30 ($^4J_{\text{CF}}$ = 8.3 Hz), 128.65 ($^3J_{\text{CF}}$ = 8.2 Hz), 121.80, 121.77, 115.99 ($^2J_{\text{CF}}$ = 21.8 Hz), 62.21, 62.08.

HRMS (EI+): m/z calcd for $\text{C}_{10}\text{H}_{10}\text{FNO}$ ($M + \text{H}$)⁺: 180.0825; found: 180.0898.

(2E)-3-(4-Chlorophenyl)acrylaldehyde O-methyl Oxime (5a)²⁷

White solid; yield: 341 mg (35%); mp 83.5–84 °C.

^1H NMR (400 MHz, CDCl_3): δ = 7.84 (d, J = 8.8 Hz, 1 H), 7.45–7.34 (m, 2 H), 7.34–7.27 (m, 2 H), 6.83–6.75 (m, 1 H), 6.72 (d, J = 16.0 Hz, 1 H), 3.94 (d, J = 15.9 Hz, 3 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 150.27, 147.72, 138.43, 137.09, 134.64, 134.62, 129.16, 128.73, 128.14, 122.62, 116.80, 62.14.

HRMS (EI+): m/z calcd for $\text{C}_{10}\text{H}_{10}\text{ClNO}$ ($M + \text{H}$)⁺: 196.0529; found: 196.0615.

(2E)-3-(4-Bromophenyl)acrylaldehyde O-Methyl Oxime (6a)²⁷

White solid; yield: 621 mg (52%); mp 93.6–94.8 °C.

^1H NMR (400 MHz, CDCl_3): δ = 7.84 (d, J = 9.4 Hz, 1 H), 7.50–7.44 (m, 2 H), 7.37–7.26 (m, 2 H), 6.83–6.77 (m, 1 H), 6.71 (d, J = 16.1 Hz, 1 H), 3.94 (d, J = 15.6 Hz, 3 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 150.26, 147.71, 137.14, 135.07, 132.11, 128.98, 128.41, 122.85, 122.74, 116.88, 62.27, 62.16.

HRMS (EI+): m/z calcd for $\text{C}_{10}\text{H}_{10}\text{BrNO}$ ($M + \text{H}$)⁺: 240.0024; found: 240.0095.

(2E)-3-(3,4-Dimethylphenyl)acrylaldehyde O-Methyl Oxime (7a)

White solid; yield: 378 mg (40%); mp 40.3–40.9 °C.

^1H NMR (400 MHz, CDCl_3): δ = 7.84 (dd, J = 8.3, 0.8 Hz, 1 H), 7.21 (d, J = 6.3 Hz, 1 H), 7.17 (t, J = 9.0 Hz, 1 H), 7.10 (d, J = 7.7 Hz, 1 H), 6.83–6.74 (m, 1 H), 6.72 (d, J = 15.5 Hz, 1 H), 3.93 (d, J = 20.2 Hz, 3 H), 2.26 (d, J = 2.5 Hz, 6 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 150.87, 148.42, 140.41, 138.87, 137.83, 137.07, 133.80, 130.20, 128.77, 128.33, 125.29, 124.52, 120.81, 115.17, 61.98, 19.91, 19.86, 19.80.

HRMS (EI+): m/z calcd for $\text{C}_{12}\text{H}_{15}\text{NO}$ ($M + \text{H}$)⁺: 190.1232; found: 190.1222.

(2E)-3-(4-Isopropylphenyl)acrylaldehyde O-Methyl Oxime (8a)

Colorless liquid; yield: 528 mg (52%).

^1H NMR (400 MHz, CDCl_3): δ = 7.87 (dd, J = 7.5, 1.7 Hz, 1 H), 7.47–7.35 (m, 2 H), 7.23 (d, J = 8.2 Hz, 2 H), 6.87–6.73 (m, 2 H), 3.96 (d, J = 18.1 Hz, 3 H), 2.96–2.89 (m, 1 H), 1.27 (d, J = 6.9 Hz, 6 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 150.82, 150.04, 138.70, 133.77, 127.69, 127.08, 127.03, 121.05, 115.43, 61.99, 34.10, 23.96.

HRMS (EI+): m/z calcd for $\text{C}_{13}\text{H}_{17}\text{NO}$ ($M + \text{H}$)⁺: 204.1388; found: 204.1384.

(2E)-3-(Naphthalen-1-yl)acrylaldehyde O-Methyl Oxime (9a)

White solid; yield: 612 mg (58%); mp 51.2–52 °C.

^1H NMR (400 MHz, CDCl_3): δ = 8.11 (d, J = 8.2 Hz, 1 H), 8.01 (d, J = 9.9 Hz, 1 H), 7.89–7.84 (m, 1 H), 7.81 (t, J = 8.0 Hz, 1 H), 7.70 (d, J = 7.2 Hz, 1 H), 7.61–7.53 (m, 2 H), 7.51 (dd, J = 8.0, 1.6 Hz, 1 H), 7.47 (dd, J = 10.5, 4.8 Hz, 1 H), 6.94–6.88 (m, 1 H), 3.95 (s, 3 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 150.76, 135.40, 133.85, 133.39, 131.13, 129.30, 128.87, 126.62, 126.15, 125.75, 124.66, 124.11, 123.39, 62.14.

HRMS (EI+): m/z calcd for $\text{C}_{14}\text{H}_{13}\text{NO}$ ($M + \text{H}$)⁺: 212.1075; found: 212.1239.

(2E)-3-(Thiophen-3-yl)acrylaldehyde O-Methyl Oxime (10a)

Colorless liquid; yield: 300 mg (36%).

¹H NMR (400 MHz, CDCl₃): δ = 7.83–7.75 (m, 1 H), 7.30 (dd, *J* = 7.8, 2.9 Hz, 2 H), 7.25 (d, *J* = 5.0 Hz, 1 H), 6.80 (d, *J* = 16.0 Hz, 1 H), 6.70–6.60 (m, 1 H), 3.90 (d, *J* = 4.7 Hz, 3 H).¹³C NMR (101 MHz, CDCl₃): δ = 150.59, 139.11, 132.45, 126.75, 124.96, 124.58, 121.90, 62.01.HRMS (EI+): *m/z* calcd for C₈H₉NOS (M + H)⁺: 168.0483; found: 168.0468.**(2E)-3-(3-Chlorophenyl)acrylaldehyde O-Methyl Oxime (11a)**

Colorless liquid; yield: 331 mg (34%).

¹H NMR (400 MHz, CDCl₃): δ = 7.92–7.83 (m, 1 H), 7.42–7.37 (m, 1 H), 7.30 (ddd, *J* = 6.4, 3.1, 1.4 Hz, 1 H), 7.28–7.22 (m, 2 H), 6.81 (dd, *J* = 16.0, 9.5 Hz, 1 H), 6.70 (d, *J* = 16.1 Hz, 1 H), 3.93 (d, *J* = 2.1 Hz, 3 H).¹³C NMR (101 MHz, CDCl₃): δ = 150.41, 150.05, 137.98, 136.78, 134.90, 130.12, 128.74, 126.94, 125.00, 123.47, 62.17.HRMS (EI+): *m/z* calcd for C₁₀H₉ClNO (M + H)⁺: 196.0529; found: 192.0615.**(2E)-3-(*m*-Tolyl)acrylaldehyde O-Methyl Oxime (12a)**

Colorless liquid; yield: 350 mg (40%).

¹H NMR (400 MHz, CDCl₃): δ = 7.84 (d, *J* = 9.1 Hz, 1 H), 7.27–7.20 (m, 3 H), 7.13–7.06 (m, 1 H), 6.81 (dd, *J* = 16.0, 9.1 Hz, 1 H), 6.73 (d, *J* = 16.0 Hz, 1 H), 3.91 (s, 3 H), 2.34 (s, 3 H).¹³C NMR (101 MHz, CDCl₃): δ = 150.65, 138.76, 138.47, 136.05, 129.73, 128.79, 127.76, 124.12, 121.78, 62.00, 21.47.HRMS (EI+): *m/z* calcd for C₁₁H₁₃NO (M + H)⁺: 176.1075; found: 176.1081.**(2E)-3-(2-Fluorophenyl)acrylaldehyde O-Methyl Oxime (13a)**

Colorless liquid; yield: 286 mg (32%).

¹H NMR (400 MHz, CDCl₃): δ = 7.87 (d, *J* = 8.4 Hz, 1 H), 7.49 (td, *J* = 7.7, 1.4 Hz, 1 H), 7.30–7.22 (m, 1 H), 7.12 (t, *J* = 7.4 Hz, 1 H), 7.05 (dd, *J* = 10.2, 8.8 Hz, 1 H), 6.95 (d, *J* = 16.0 Hz, 1 H), 6.92–6.85 (m, 1 H), 3.93 (s, 3 H).¹³C NMR (101 MHz, CDCl₃): δ = 160.59 (¹*J*_{CF} = 252.4 Hz), 150.67 (s), 130.74 (³*J*_{CF} = 3.6 Hz), 130.24, 130.16, 127.53, 127.50, 124.48, 124.44, 124.30 (⁴*J*_{CF} = 5.45 Hz), 124.16, 124.04, 116.08 (²*J*_{CF} = 22.12 Hz), 62.12.HRMS (EI+): *m/z* calcd for C₁₀H₁₀FNO (M + H)⁺: 180.0825; found: 180.0898.**(2E)-3-(4-Nitrophenyl)acrylaldehyde O-Methyl Oxime (14a)**

White solid; yield: 618 mg (60%); mp 121.6–122.4 °C.

¹H NMR (400 MHz, CDCl₃): δ = 8.23–8.18 (m, 2 H), 7.88 (d, *J* = 9.6 Hz, 1 H), 7.57 (d, *J* = 8.8 Hz, 2 H), 6.96 (dd, *J* = 16.1, 9.6 Hz, 1 H), 6.83 (d, *J* = 16.1 Hz, 1 H), 3.96 (s, 3 H).¹³C NMR (101 MHz, CDCl₃): δ = 149.57, 147.58, 142.42, 135.43, 127.43, 126.48, 124.31, 62.41.HRMS (EI+): *m/z* calcd for C₁₀H₁₀N₂O₃ (M + H)⁺: 207.0770; found: 207.0850.**4-[(1E)-3-(Methoxyimino)prop-1-en-1-yl]benzonitrile (15a)**

White solid; yield: 502 mg (54%); mp 112–112.9 °C.

¹H NMR (400 MHz, CDCl₃): δ = 7.87 (d, *J* = 9.6 Hz, 1 H), 7.63 (d, *J* = 8.4 Hz, 2 H), 7.51 (d, *J* = 8.3 Hz, 2 H), 6.91 (dd, *J* = 16.1, 9.6 Hz, 1 H), 6.77 (d, *J* = 16.1 Hz, 1 H), 3.95 (s, 3 H).¹³C NMR (101 MHz, CDCl₃): δ = 149.67, 140.50, 135.97, 132.71, 127.33, 125.67, 118.81, 111.92, 62.36.HRMS (EI+): *m/z* calcd for C₁₁H₁₀N₂O (M + H)⁺: 187.0871; found: 187.0946.**(2E)-3-(Naphthalen-2-yl)acrylaldehyde O-Methyl Oxime (16a)**

White solid; yield: 443 mg (42%); mp 88.6–89.6 °C.

¹H NMR (400 MHz, CDCl₃): δ = 7.94–7.89 (m, 1 H), 7.80 (d, *J* = 8.9 Hz, 4 H), 7.62 (dd, *J* = 8.6, 1.4 Hz, 1 H), 7.47 (dd, *J* = 5.9, 3.3 Hz, 2 H), 6.94 (d, *J* = 4.9 Hz, 2 H), 3.94 (s, 3 H).¹³C NMR (101 MHz, CDCl₃): δ = 150.68, 138.69, 133.66, 133.59, 128.69, 128.36, 127.88, 127.85, 126.69, 123.31, 122.30, 62.11.HRMS (EI+): *m/z* calcd for C₁₄H₁₃NO (M + H)⁺: 212.1075; found: 212.1239.**(2E)-3-[4-(Trifluoromethyl)phenyl]acrylaldehyde O-Methyl Oxime (17a)**

White solid; yield: 550 mg (48%); mp 42.8–43.6 °C.

¹H NMR (400 MHz, CDCl₃): δ = 7.87 (d, *J* = 9.4 Hz, 1 H), 7.60 (d, *J* = 8.3 Hz, 2 H), 7.52 (d, *J* = 8.3 Hz, 2 H), 6.90 (dd, *J* = 16.1, 9.4 Hz, 1 H), 6.79 (d, *J* = 16.1 Hz, 1 H), 3.94 (s, 3 H).¹³C NMR (101 MHz, CDCl₃): δ = 149.96, 139.53, 136.59, 130.48 (²*J*_{CF} = 32.7 Hz), 127.09, 125.90 (³*J*_{CF} = 3.8 Hz), 125.49, 124.55, 62.25.HRMS (EI+): *m/z* calcd for C₁₁H₁₀F₃NO (M + H)⁺: 230.0793; found: 230.0885.**(2E)-3-(Pyridin-2-yl)acrylaldehyde O-methyl Oxime (18a)**

Colorless liquid; yield: 259 mg (32%).

¹H NMR (400 MHz, CDCl₃): δ = 8.61–8.56 (m, 1 H), 7.90 (d, *J* = 10.1 Hz, 1 H), 7.65 (td, *J* = 7.8, 1.6 Hz, 1 H), 7.37 (d, *J* = 7.9 Hz, 1 H), 7.29–7.20 (m, 1 H), 7.19–7.14 (m, 1 H), 6.86 (d, *J* = 15.9 Hz, 1 H), 3.94 (s, 3 H).¹³C NMR (101 MHz, CDCl₃): δ = 154.60, 149.94, 137.79, 136.59, 125.93, 122.95, 122.05, 100.07, 62.16.HRMS (EI+): *m/z* calcd for C₉H₁₀N₂O (M + H)⁺: 163.0871; found: 163.0918.**(2E)-Oct-2-enal O-Methyl Oxime (19a)**

Colorless liquid; yield: 589 mg (76%).

¹H NMR (400 MHz, CDCl₃): δ = 7.67 (d, *J* = 9.5 Hz, 1 H), 6.14–6.07 (m, 1 H), 6.05–5.96 (m, 1 H), 3.86 (d, *J* = 10.7 Hz, 3 H), 2.22–2.12 (m, 2 H), 1.46–1.39 (m, 2 H), 1.34–1.25 (m, 4 H), 0.89 (t, *J* = 6.9 Hz, 3 H).¹³C NMR (101 MHz, CDCl₃): δ = 150.65, 146.60, 142.75, 139.78, 123.82, 121.81, 61.67, 32.85, 31.40, 29.13, 28.39, 28.16, 22.57, 14.09.HRMS (EI+): *m/z* calcd for C₉H₁₇NO (M + H)⁺: 156.1388; found: 156.1382.**(2E)-3-(Benzo[d][1,3]dioxol-5-yl)acrylaldehyde O-Methyl Oxime (20a)**

White solid; yield: 595 mg (58%); mp 81.1–81.9 °C.

¹H NMR (400 MHz, CDCl₃): δ = 7.83 (d, *J* = 8.7 Hz, 1 H), 6.97 (d, *J* = 1.4 Hz, 1 H), 6.88 (dd, *J* = 8.0, 1.5 Hz, 1 H), 6.78 (d, *J* = 8.0 Hz, 1 H), 6.74–6.59 (m, 2 H), 5.98 (d, *J* = 2.4 Hz, 2 H), 3.93 (d, *J* = 15.9 Hz, 3 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 150.70, 148.44, 148.39, 138.35, 130.68, 122.53, 120.22, 108.63, 105.85, 101.46, 62.00.

HRMS (EI+): m/z calcd for $\text{C}_{11}\text{H}_{11}\text{NO}_3$ ($M + \text{H}$) $^+$: 206.0817; found: 206.0896.

6,6-Dimethylbicyclo[3.1.1]hept-2-ene-2-carbaldehyde O-Methyl Oxime (21a)

Colorless liquid; yield: 376 mg (42%).

^1H NMR (400 MHz, CDCl_3): δ = 7.66 (s, 1 H), 5.83 (t, J = 3.3 Hz, 1 H), 3.85 (s, 3 H), 2.83 (td, J = 5.7, 1.1 Hz, 1 H), 2.48–2.43 (m, 1 H), 2.43–2.33 (m, 2 H), 2.17–2.11 (m, 1 H), 1.33 (s, 3 H), 1.16 (d, J = 9.0 Hz, 1 H), 0.82 (s, 3 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 150.17, 143.02, 131.14, 61.68, 40.84, 40.50, 37.76, 32.47, 31.26, 26.12, 20.99.

HRMS (EI+): m/z calcd for $\text{C}_{11}\text{H}_{17}\text{NO}$ ($M + \text{H}$) $^+$: 180.1388; found: 180.1389.

Benzaldehyde O-Methyl Oxime (22a)

Colorless liquid; yield: 621 mg (92%).

^1H NMR (400 MHz, CDCl_3): δ = 8.05 (s, 1 H), 7.60–7.52 (m, 2 H), 7.38–7.32 (m, 3 H), 3.96 (s, 3 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 148.60, 132.29, 129.85, 128.75, 127.08, 62.03.

Data were consistent with those previously reported.²⁸

1-Phenylethan-1-one O-Methyl Oxime (23a)

Colorless liquid; yield: 670 mg (90%).

^1H NMR (400 MHz, CDCl_3): δ = 7.68–7.58 (m, 2H), 7.40–7.29 (m, 3H), 3.99 (d, J = 1.3 Hz, 3H), 2.22 (d, J = 0.9 Hz, 3H).

^{13}C NMR (101 MHz, CDCl_3): δ = 154.67, 136.72, 129.09, 128.46, 126.10, 61.95, 12.67.

Data were consistent with those previously reported.²⁹

Homoallylic Amines 1c–27c; General Procedure

The respective α,β -unsaturated oxime ether (0.5 mmol) was added to a THF solution of the corresponding organozinc reagent (1 mmol) at -20°C under a N_2 atmosphere. The mixture was stirred for 10 h. When the reaction was complete, the temperature of the mixture was raised to r.t. and the mixture was diluted with H_2O . The resulting mixture was extracted with EtOAc ($3 \times 10\text{ mL}$). The combined organic extracts were washed with brine and dried (anhyd Na_2SO_4). The solvent was removed by evaporation under reduced pressure, and the residue was purified by column chromatography on basic alumina (eluting with 20:1 PE/ EtOAc) to give the desired homoallylic amine.

(E)-O-Methyl-N-(1-phenylhexa-1,5-dien-3-yl)hydroxylamine (1c)

Yellow liquid; yield: 88 mg (87%).

^1H NMR (400 MHz, CDCl_3): δ = 7.40–7.36 (m, 2 H), 7.33–7.28 (m, 2 H), 7.25–7.20 (m, 1 H), 6.56 (d, J = 16.0 Hz, 1 H), 6.17–6.11 (m, 1 H), 5.87–5.75 (m, 1 H), 5.17–5.07 (m, 2 H), 3.67–3.62 (m, 1 H), 3.55 (s, 3 H), 2.35 (td, J = 9.3, 7.9, 5.1, 1.2 Hz, 2 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 137.04, 134.61, 132.41, 129.67, 128.67, 127.70, 126.55, 117.97, 63.00, 62.68, 37.11.

HRMS (EI+): m/z calcd for $\text{C}_{13}\text{H}_{17}\text{NO}$ ($M + \text{H}$) $^+$: 204.1388; found: 204.1387.

(E)-O-Methyl-N-[1-(p-tolyl)hexa-1,5-dien-3-yl]hydroxylamine (2c)

Yellow liquid; yield: 89 mg (82%).

^1H NMR (400 MHz, CDCl_3): δ = 7.27 (d, J = 8.1 Hz, 2 H), 7.11 (d, J = 7.9 Hz, 2 H), 6.52 (d, J = 15.9 Hz, 1 H), 6.11–6.05 (m, 1 H), 5.86–5.74 (m, 1 H), 5.61 (s, 1 H), 5.17–5.06 (m, 2 H), 3.65–3.60 (m, 1 H), 3.54 (s, 3 H), 2.43–2.27 (m, 5 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 137.45, 134.66, 134.21, 132.25, 129.31, 128.50, 126.40, 117.81, 63.00, 62.61, 37.11, 21.27.

HRMS (EI+): m/z calcd for $\text{C}_{14}\text{H}_{19}\text{NO}$ ($M + \text{H}$) $^+$: 218.1545; found: 218.1556.

(E)-N-[1-(4-Methoxyphenyl)hexa-1,5-dien-3-yl]-O-methylhydroxylamine (3c)

Yellow liquid; yield: 71 mg (61%).

^1H NMR (400 MHz, CDCl_3): δ = 7.34–7.29 (m, 2 H), 6.86–6.81 (m, 2 H), 6.50 (d, J = 15.9 Hz, 1 H), 6.02–5.96 (m, 1 H), 5.87–5.75 (m, 1 H), 5.17–5.06 (m, 2 H), 3.79 (s, 3 H), 3.64–3.59 (m, 1 H), 3.54 (s, 3 H), 2.44–2.27 (m, 2 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 159.31, 134.73, 131.87, 129.79, 127.67, 127.30, 117.77, 114.05, 63.05, 62.62, 55.38, 37.17.

HRMS (EI+): m/z calcd for $\text{C}_{14}\text{H}_{19}\text{NO}_2\text{Na}$ ($M + \text{Na}$) $^+$: 256.1313; found: 256.1493.

(E)-N-[1-(4-Fluorophenyl)hexa-1,5-dien-3-yl]-O-methylhydroxylamine (4c)

Yellow liquid; yield: 92 mg (84%).

^1H NMR (400 MHz, CDCl_3): δ = 7.37–7.29 (m, 2 H), 7.03–6.95 (m, 2 H), 6.52 (d, J = 16.0 Hz, 1 H), 6.08–6.02 (m, 1 H), 5.86–5.74 (m, 1 H), 5.62 (s, 1 H), 5.13 (dt, J = 10.2, 5.2 Hz, 2 H), 3.66–3.60 (m, 1 H), 3.54 (s, 3 H), 2.41–2.30 (m, 2 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 162.39 ($^1J_{\text{CF}}$ = 247.7 Hz), 134.50, 133.16 ($^4J_{\text{CF}}$ = 3.3 Hz), 131.13, 129.43 ($^3J_{\text{CF}}$ = 2.1 Hz), 128.02, 127.94, 117.95, 115.51 ($^2J_{\text{CF}}$ = 21.6 Hz), 62.84, 62.59, 37.03.

HRMS (EI+): m/z calcd for $\text{C}_{13}\text{H}_{16}\text{FNO}$ ($M + \text{H}$) $^+$: 222.1294; found: 222.1267.

(E)-N-[1-(4-Chlorophenyl)hexa-1,5-dien-3-yl]-O-methylhydroxylamine (5c)

Yellow liquid; yield: 97 mg (82%).

^1H NMR (400 MHz, CDCl_3): δ = 7.32–7.26 (m, 4 H), 6.51 (d, J = 16.0 Hz, 1 H), 6.15–6.09 (m, 1 H), 5.86–5.74 (m, 1 H), 5.62 (s, 1 H), 5.17–5.08 (m, 2 H), 3.64 (q, J = 7.1 Hz, 1 H), 3.54 (s, 3 H), 2.43–2.28 (m, 2 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 135.53, 134.44, 133.29, 131.10, 130.47, 128.82, 127.73, 118.10, 62.82, 62.66, 37.01.

HRMS (EI+): m/z calcd for $\text{C}_{13}\text{H}_{16}\text{ClNO}$ ($M + \text{H}$) $^+$: 238.0999; found: 238.1014.

(E)-N-[1-(4-Bromophenyl)hexa-1,5-dien-3-yl]-O-methylhydroxylamine (6c)

Yellow liquid; yield: 113 mg (81%).

^1H NMR (400 MHz, CDCl_3): δ = 7.44–7.41 (m, 2 H), 7.26–7.22 (m, 2 H), 6.49 (d, J = 16.0 Hz, 1 H), 6.17–6.11 (m, 1 H), 5.85–5.73 (m, 1 H), 5.62 (s, 1 H), 5.17–5.09 (m, 2 H), 3.63 (q, J = 6.8 Hz, 1 H), 3.53 (d, J = 6.7 Hz, 3 H), 2.43–2.27 (m, 2 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 135.97, 134.42, 131.76, 131.12, 130.64, 128.05, 121.42, 118.10, 62.81, 62.65, 36.98.

HRMS (EI⁺): *m/z* calcd for C₁₃H₁₆BrNONa (M + Na)⁺: 304.0313; found: 304.0318.

(E)-N-[1-(3,4-Dimethylphenyl)hexa-1,5-dien-3-yl]-O-methylhydroxylamine (7c)

Yellow liquid; yield: 62 mg (54%).

¹H NMR (400 MHz, CDCl₃): δ = 7.16 (s, 1 H), 7.12 (d, *J* = 7.9 Hz, 1 H), 7.06 (d, *J* = 7.7 Hz, 1 H), 6.50 (d, *J* = 15.9 Hz, 1 H), 6.10–6.04 (m, 1 H), 5.86–5.74 (m, 1 H), 5.11 (t, *J* = 14.8 Hz, 2 H), 3.65–3.60 (m, 1 H), 3.54 (s, 3 H), 2.43–2.28 (m, 2 H), 2.24 (d, *J* = 3.1 Hz, 6 H).

¹³C NMR (101 MHz, CDCl₃): δ = 136.71, 136.20, 134.71, 134.67, 132.40, 129.91, 128.34, 127.76, 124.01, 117.80, 63.09, 62.66, 37.15, 19.88, 19.62.

HRMS (EI⁺): *m/z* calcd for C₁₅H₂₁NONa (M + Na)⁺: 254.1521; found: 254.1531.

(E)-N-[1-(4-Isopropylphenyl)hexa-1,5-dien-3-yl]-O-methylhydroxylamine (8c)

Yellow liquid; yield: 70 mg (57%).

¹H NMR (400 MHz, CDCl₃): δ = 7.31 (d, *J* = 8.1 Hz, 2 H), 7.17 (d, *J* = 8.1 Hz, 2 H), 6.53 (d, *J* = 15.9 Hz, 1 H), 6.12–6.06 (m, 1 H), 5.86–5.74 (m, 1 H), 5.61 (s, 1 H), 5.15–5.08 (m, 2 H), 3.66–3.61 (m, 1 H), 3.54 (s, 3 H), 2.92–2.85 (m, 1 H), 2.44–2.27 (m, 2 H), 1.24 (d, *J* = 6.9 Hz, 6 H).

¹³C NMR (101 MHz, CDCl₃): δ = 148.58, 134.68, 132.32, 128.66, 126.73, 126.53, 117.86, 63.10, 62.68, 37.16, 33.99, 24.07.

HRMS (EI⁺): *m/z* calcd for C₁₆H₂₃NO (M + H)⁺: 246.1858; found: 246.1865.

(E)-O-Methyl-N-[1-(naphthalen-1-yl)hexa-1,5-dien-3-yl]hydroxylamine (9c)

Yellow liquid; yield: 67 mg (53%).

¹H NMR (400 MHz, CDCl₃): δ = 8.11 (d, *J* = 7.9 Hz, 1 H), 7.86–7.81 (m, 1 H), 7.77 (d, *J* = 8.2 Hz, 1 H), 7.58 (d, *J* = 7.1 Hz, 1 H), 7.54–7.46 (m, 2 H), 7.46–7.41 (m, 1 H), 7.33–7.23 (m, 1 H), 6.19–6.13 (m, 1 H), 5.92–5.81 (m, 1 H), 5.70 (s, 1 H), 5.20–5.13 (m, 2 H), 3.81–3.76 (m, 1 H), 3.59 (s, 3 H), 2.48–2.38 (m, 2 H).

¹³C NMR (101 MHz, CDCl₃): δ = 134.93, 134.63, 133.72, 133.01, 131.28, 129.69, 128.63, 128.02, 126.13, 125.88, 125.74, 124.03, 123.99, 118.06, 63.18, 62.76, 37.17.

HRMS (EI⁺): *m/z* calcd for C₁₇H₁₉NONa (M + Na)⁺: 276.1364; found: 276.1342.

(E)-O-Methyl-N-[1-(thiophen-3-yl)hexa-1,5-dien-3-yl]hydroxylamine (10c)

Yellow liquid; yield: 62 mg (60%).

¹H NMR (400 MHz, CDCl₃): δ = 7.26–7.24 (m, 1 H), 7.21 (dd, *J* = 5.0, 1.1 Hz, 1 H), 7.15–7.12 (m, 1 H), 6.57 (d, *J* = 15.9 Hz, 1 H), 6.01–5.95 (m, 1 H), 5.85–5.74 (m, 1 H), 5.57 (s, 1 H), 5.11–5.08 (m, 2 H), 3.63–3.57 (m, 1 H), 3.55–3.52 (m, 3 H), 2.42–2.26 (m, 2 H).

¹³C NMR (101 MHz, CDCl₃): δ = 139.67, 134.61, 129.48, 126.60, 126.09, 125.13, 122.19, 117.91, 62.92, 62.64, 37.07.

HRMS (EI⁺): *m/z* calcd for C₁₁H₁₅NOS (M + H)⁺: 210.0953; found: 210.0962.

(E)-N-[1-(3-Chlorophenyl)hexa-1,5-dien-3-yl]-O-methylhydroxylamine (11c)

Yellow liquid; yield: 93 mg (79%).

¹H NMR (400 MHz, CDCl₃): δ = 7.39–7.32 (m, 1 H), 7.25–7.14 (m, 3 H), 6.50 (d, *J* = 16.0 Hz, 1 H), 6.20–6.11 (m, 1 H), 5.86–5.74 (m, 1 H), 5.62 (s, 1 H), 5.19–5.09 (m, 2 H), 3.73–3.62 (m, 1 H), 3.55 (d, *J* = 7.5 Hz, 3 H), 2.45–2.27 (m, 2 H).

¹³C NMR (101 MHz, CDCl₃): δ = 138.93, 134.62, 134.37, 131.45, 130.96, 129.88, 127.61, 126.44, 124.77, 118.15, 62.77, 62.67, 36.97.

HRMS (EI⁺): *m/z* calcd for C₁₃H₁₆ClNO (M + H)⁺: 238.0999; found: 238.1016.

(E)-O-Methyl-N-[1-(*m*-tolyl)hexa-1,5-dien-3-yl]hydroxylamine (12c)

Yellow liquid; yield: 73 mg (68%).

¹H NMR (400 MHz, CDCl₃): δ = 7.23–7.15 (m, 3 H), 7.05 (d, *J* = 5.6 Hz, 1 H), 6.53 (d, *J* = 15.9 Hz, 1 H), 6.15–6.09 (m, 1 H), 5.86–5.75 (m, 1 H), 5.62 (s, 1 H), 5.17–5.07 (m, 2 H), 3.66–3.61 (m, 1 H), 3.55 (s, 3 H), 2.44–2.29 (m, 5 H).

¹³C NMR (101 MHz, CDCl₃): δ = 138.20, 136.97, 134.65, 132.49, 129.43, 128.56, 128.48, 127.24, 123.71, 117.90, 63.03, 62.68, 37.12, 21.52.

HRMS (EI⁺): *m/z* calcd for C₁₄H₁₉NO (M + H)⁺: 218.1545; found: 218.1556.

(E)-N-[1-(2-Fluorophenyl)hexa-1,5-dien-3-yl]-O-methylhydroxylamine (13c)

Yellow liquid; yield: 91 mg (83%).

¹H NMR (400 MHz, CDCl₃): δ = 7.45 (td, *J* = 7.7, 1.7 Hz, 1 H), 7.23–7.16 (m, 1 H), 7.11–6.99 (m, 2 H), 6.72 (d, *J* = 16.1 Hz, 1 H), 6.26–6.20 (m, 1 H), 5.87–5.76 (m, 1 H), 5.64 (s, 1 H), 5.17–5.08 (m, 2 H), 3.69–3.64 (m, 1 H), 3.55 (s, 3 H), 2.44–2.29 (m, 2 H).

¹³C NMR (101 MHz, CDCl₃): δ = 160.34 (¹*J*_{CF} = 250.3 Hz), 134.48, 132.46, 132.41, 128.91 (²*J*_{CF} = 8.4 Hz), 127.51, 127.48, 124.72, 124.68, 124.16 (⁴*J*_{CF} = 3.6 Hz), 118.05 (s), 115.83 (²*J*_{CF} = 22.2 Hz), 63.16, 62.65, 37.02.

HRMS (EI⁺): *m/z* calcd for C₁₃H₁₆FNO (M + H)⁺: 222.1294; found: 222.1267.

(E)-O-Methyl-N-[1-(4-nitrophenyl)hexa-1,5-dien-3-yl]hydroxylamine (14c)

Yellow liquid; yield: 38 mg (31%).

¹H NMR (400 MHz, CDCl₃): δ = 8.18 (d, *J* = 8.8 Hz, 2 H), 7.51 (d, *J* = 8.7 Hz, 2 H), 6.63 (d, *J* = 16.0 Hz, 1 H), 6.39–6.33 (m, 1 H), 5.85–5.75 (m, 1 H), 5.65 (s, 1 H), 5.20–5.10 (m, 2 H), 3.72–3.67 (m, 1 H), 3.55 (s, 3 H), 2.45–2.30 (m, 2 H).

¹³C NMR (101 MHz, CDCl₃): δ = 147.06, 143.51, 135.18, 134.04, 130.14, 127.03, 124.13, 118.49, 62.66, 62.64, 36.81.

HRMS (EI⁺): *m/z* calcd for C₁₃H₁₆N₂O₃ (M + H)⁺: 249.1239; found: 249.1215.

(E)-O-Methyl-N-[1-(naphthalen-2-yl)hexa-1,5-dien-3-yl]hydroxylamine (16c)

Yellow liquid; yield: 78 mg (62%).

¹H NMR (400 MHz, CDCl₃): δ = 7.81–7.76 (m, 3 H), 7.72 (s, 1 H), 7.60 (d, *J* = 8.5 Hz, 1 H), 7.48–7.40 (m, 2 H), 6.72 (d, *J* = 15.9 Hz, 1 H), 6.30–6.24 (m, 1 H), 5.89–5.79 (m, 1 H), 5.19–5.09 (m, 2 H), 3.73–3.68 (m, 1 H), 3.56 (t, *J* = 5.3 Hz, 3 H), 2.48–2.33 (m, 2 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 134.61, 134.49, 133.70, 133.11, 132.49, 130.08, 128.29, 128.08, 127.78, 126.45, 126.37, 125.96, 123.70, 118.01, 63.06, 62.69, 37.13.

HRMS (EI⁺): m/z calcd for $\text{C}_{17}\text{H}_{19}\text{NONa}$ ($\text{M} + \text{Na}$)⁺: 276.1364; found: 276.1342.

(E)-O-Methyl-N-{1-[4-(trifluoromethyl)phenyl]hexa-1,5-dien-3-yl}hydroxylamine (17c)

Yellow liquid; yield: 111 mg (82%).

^1H NMR (400 MHz, CDCl_3): δ = 7.56 (d, J = 8.2 Hz, 2 H), 7.47 (d, J = 8.2 Hz, 2 H), 6.59 (d, J = 16.0 Hz, 1 H), 6.28–6.23 (m, 1 H), 5.87–5.74 (m, 1 H), 5.64 (s, 1 H), 5.18–5.10 (m, 2 H), 3.67 (q, J = 6.9 Hz, 1 H), 3.55 (s, 3 H), 2.44–2.29 (m, 2 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 140.53, 134.27, 132.70, 130.92, 126.67, 125.63 ($J_{\text{C-F}}$ = 3.9 Hz), 118.25, 62.75, 62.66, 36.93.

HRMS (EI⁺): m/z calcd for $\text{C}_{14}\text{H}_{16}\text{F}_3\text{NO}$ ($\text{M} + \text{H}$)⁺: 272.1262; found: 272.1272.

(E)-O-Methyl-N-[1-(pyridin-2-yl)hexa-1,5-dien-3-yl]hydroxylamine (18c)

Yellow liquid; yield: 84 mg (83%).

^1H NMR (400 MHz, CDCl_3): δ = 8.55 (ddd, J = 4.8, 1.7, 0.8 Hz, 1 H), 7.62 (td, J = 7.7, 1.8 Hz, 1 H), 7.31–7.28 (m, 1 H), 7.12 (ddd, J = 7.5, 4.8, 1.1 Hz, 1 H), 6.66 (d, J = 3.6 Hz, 2 H), 5.88–5.76 (m, 1 H), 5.19–5.08 (m, 2 H), 3.74–3.69 (m, 1 H), 3.55 (s, 3 H), 2.45–2.31 (m, 2 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 155.37, 149.59, 136.53, 134.42, 131.96, 122.22, 121.60, 118.05, 62.53, 62.41, 36.82.

HRMS (EI⁺): m/z calcd for $\text{C}_{12}\text{H}_{16}\text{N}_2\text{O}$ ($\text{M} + \text{H}$)⁺: 205.1341; found: 205.1328.

(E)-O-Methyl-N-(undeca-1,5-dien-4-yl)hydroxylamine (19c)

Yellow liquid; yield: 89 mg (91%).

^1H NMR (400 MHz, CDCl_3): δ = 5.83–5.69 (m, 1 H), 5.67–5.51 (m, 1 H), 5.37–5.22 (m, 1 H), 5.14–4.99 (m, 2 H), 3.57–3.47 (m, 3 H), 3.46–3.40 (m, 1 H), 2.36–2.12 (m, 2 H), 2.11–1.93 (m, 2 H), 1.41–1.20 (m, 6 H), 0.88 (t, J = 6.8 Hz, 3 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 135.04, 134.24, 129.46, 117.42, 62.83, 62.56, 37.17, 32.52, 31.45, 28.98, 22.63, 14.18.

HRMS (EI⁺): m/z calcd for $\text{C}_{12}\text{H}_{23}\text{NO}$ ($\text{M} + \text{H}$)⁺: 204.1388; found: 204.1387.

(E)-N-[1-(Benzo[d][1,3]dioxol-5-yl)hexa-1,5-dien-3-yl]-O-methylhydroxylamine (20c)

Yellow liquid; yield: 84 mg (68%).

^1H NMR (400 MHz, CDCl_3): δ = 6.92 (d, J = 1.6 Hz, 1 H), 6.78 (dt, J = 22.5, 4.8 Hz, 2 H), 6.46 (d, J = 15.9 Hz, 1 H), 5.99–5.92 (m, 3 H), 5.85–5.74 (m, 1 H), 5.15–5.08 (m, 2 H), 3.63–3.58 (m, 1 H), 3.54 (s, 3 H), 2.42–2.26 (m, 2 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 148.11, 147.31, 134.64, 132.01, 131.52, 127.83, 121.20, 117.89, 108.36, 105.82, 101.17, 62.94, 62.65, 37.15.

HRMS (EI⁺): m/z calcd for $\text{C}_{14}\text{H}_{17}\text{NO}_3\text{Na}$ ($\text{M} + \text{Na}$)⁺: 270.1106; found: 270.1073.

(E)-O-Methyl-N-(4-methyl-1-phenylhexa-1,5-dien-3-yl)hydroxylamine (22c)

Yellow liquid; yield: 64 mg (59%).

^1H NMR (400 MHz, CDCl_3): δ = 7.38 (dd, J = 9.4, 7.9 Hz, 2 H), 7.34–7.28 (m, 2 H), 7.25–7.20 (m, 1 H), 6.55 (d, J = 15.9 Hz, 1 H), 6.07 (dt, J = 15.9, 8.2 Hz, 1 H), 5.91–5.79 (m, 1 H), 5.61 (s, 1 H), 5.15–5.05 (m, 2 H), 3.57–3.50 (m, 4 H), 2.67–2.59 (m, 1 H), 1.06 (d, J = 7.0 Hz, 3 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 139.93, 137.07, 133.55, 128.64, 127.66, 127.11, 126.55, 115.81, 67.79, 62.27, 39.24, 16.69.

HRMS (EI⁺): m/z calcd for $\text{C}_{14}\text{H}_{19}\text{NO}$ ($\text{M} + \text{H}$)⁺: 218.1545; found: 218.1546.

(E)-O-Methyl-N-(5-methyl-1-phenylhexa-1,5-dien-3-yl)hydroxylamine (23c)

Yellow liquid; yield: 76 mg (70%).

^1H NMR (400 MHz, CDCl_3): δ = 7.39 (dd, J = 5.2, 3.4 Hz, 2 H), 7.35–7.28 (m, 2 H), 7.25–7.19 (m, 1 H), 6.57 (d, J = 15.9 Hz, 1 H), 6.18–6.12 (m, 1 H), 5.63 (s, 1 H), 4.87–4.76 (m, 2 H), 3.78–3.70 (m, 1 H), 3.54 (s, 3 H), 2.40–2.34 (m, 1 H), 2.22–2.17 (m, 1 H), 1.77 (s, 3 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 141.95, 137.14, 132.01, 130.42, 128.65, 127.62, 126.52, 113.63, 62.74, 61.30, 41.12, 22.69.

HRMS (EI⁺): m/z calcd for $\text{C}_{14}\text{H}_{19}\text{NO}$ ($\text{M} + \text{H}$)⁺: 218.1545; found: 218.1545.

(E)-N-(4,4-Dimethyl-1-phenylhexa-1,5-dien-3-yl)-O-methylhydroxylamine (26c)

Yellow liquid; yield: 74 mg (64%).

^1H NMR (400 MHz, CDCl_3): δ = 7.43–7.39 (m, 2 H), 7.32 (t, J = 7.5 Hz, 2 H), 7.25–7.21 (m, 1 H), 6.55 (d, J = 15.9 Hz, 1 H), 6.18 (dd, J = 15.9, 8.7 Hz, 1 H), 5.87 (dd, J = 17.4, 10.8 Hz, 1 H), 5.11–5.03 (m, 2 H), 3.48 (s, 3 H), 3.31 (d, J = 8.7 Hz, 1 H), 1.05 (s, 6 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 145.71, 137.32, 133.67, 128.65, 128.31, 127.52, 126.57, 113.39, 71.42, 62.30, 39.81, 26.39, 21.92.

HRMS (EI⁺): m/z calcd for $\text{C}_{15}\text{H}_{21}\text{NONa}$ ($\text{M} + \text{Na}$)⁺: 254.1521; found: 254.1531.

Homoallyl Primary Amine; (E)-1-Phenylhexa-1,5-dien-3-amine (1d)³⁰

To a stirred solution of **1c** (101 mg, 0.5 mmol) in a mixture of AcOH/THF (3:1) (4 mL) was added Zn powder (1.3 g, 20 mmol) in several portions at r.t., and the stirring was continued for 4 h at 50 °C. The mixture was cooled to r.t. and adjusted to pH 10 with concd NH_4OH . Inorganic material was removed by filtration through a Celite pad. The filtrate was extracted with CHCl_3 (3 × 10 mL), the combined organic extracts were dried (Na_2SO_4), and concentrated in vacuo. The residue was purified by column chromatography on basic alumina (eluting with 1:10 EtOH/EtOAc) to give the desired product **1d**; yellow liquid; yield: 58 mg (67%).

^1H NMR (400 MHz, CDCl_3): δ = 7.36 (d, J = 7.3 Hz, 2 H), 7.29 (t, J = 7.5 Hz, 2 H), 7.22 (t, J = 7.2 Hz, 1 H), 6.54 (d, J = 15.9 Hz, 1 H), 6.19 (dd, J = 15.9, 7.1 Hz, 1 H), 5.83–5.71 (m, 1 H), 5.13 (t, J = 12.8 Hz, 2 H), 3.68–3.57 (m, 3 H), 2.44–2.30 (m, 2 H).

^{13}C NMR (101 MHz, CDCl_3): δ = 136.79, 134.21, 131.64, 130.81, 128.66, 127.71, 126.54, 118.47, 53.39, 41.35.

HRMS (EI⁺): m/z calcd for $\text{C}_{12}\text{H}_{16}\text{N}$ ($\text{M} + \text{H}$)⁺: 174.1283; found: 174.1284.

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Supporting Information

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